

12.36

EE25BTECH11022 - sankeerthan

Question: Given the regression model: $Y_i = \alpha + \beta x_i + \epsilon_i$ with data:

Year	Production
2001	30
2002	35
2003	36
2004	32
2005	37
2006	40

(1)

Solution: Let

$$x_i = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{pmatrix}, Y = \begin{pmatrix} 30 \\ 35 \\ 36 \\ 32 \\ 37 \\ 40 \end{pmatrix} \quad (2)$$

Let

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \end{pmatrix} \quad (3)$$

$$X^T X = \begin{pmatrix} 6 & 21 \\ 21 & 91 \end{pmatrix}, X^T Y = \begin{pmatrix} 210 \\ 761 \end{pmatrix} \quad (4)$$

$$(X^T X)^{-1} = \begin{pmatrix} 0.87 & -0.20 \\ -0.20 & 0.06 \end{pmatrix} \quad (5)$$

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = (X^T X)^{-1} (X^T Y) \quad (6)$$

$$= \begin{pmatrix} 0.87 & -0.20 \\ -0.20 & 0.06 \end{pmatrix} \quad (7)$$

$$= \begin{pmatrix} 210 \\ 761 \end{pmatrix} = \begin{pmatrix} 29.80 \\ 1.49 \end{pmatrix} \quad (8)$$

Thus, $\alpha + \beta = 31.29$

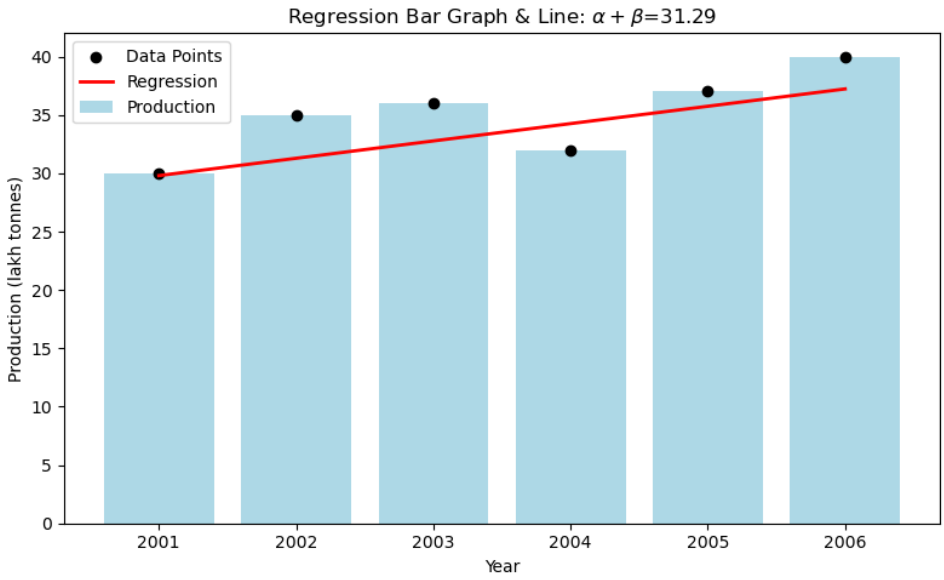


Fig. 0.1